
Prompt-Routed Multimodal Generation with a Shared World–Action Head

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Abstract

World action models usually predict RGB futures and robot actions, while depth, semantics, and surface geometry are produced by separate networks or modality-specific heads. We present CoGen, a prompt-routed extension of Action Images in which RGB video, metric depth, dense scene-role segmentation, surface normals, and end-effector actions share one generative output space. Each signal is represented as an RGB-like video, encoded by the same frozen video VAE, packed into a common latent sequence, and predicted by the same diffusion-transformer output projection under one masked flow-matching loss. A task template determines the sequence layout and is serialized into the language prompt with atomic modality tags; an independent conditioning plan determines which segments are observed and which are generated. Our principal training configuration mixes video+action, video+depth, video+segmentation, and video+normal with probabilities 0.4/0.2/0.2/0.2. A preliminary pipeline diagnostic at step 4,500 shows that this single checkpoint produces all four requested auxiliary streams on the same held-in RLBench episode: 5.4 cm median action-position error, 0.160 depth AbsRel, 0.373 scene-role macro-mIoU, and 0.904 normal cosine similarity in the joint-generation regime. We distinguish these diagnostic measurements from a completed benchmark: held-out evaluation, closed-loop evaluation, and joint depth–action templates remain unreported. We also make explicit that the prompt tag and the target modality’s clean first-frame anchor co-specify the route; prompt-only routing is not claimed by the present training distribution.

1 Introduction

World action models (WAMs) learn a policy together with a predictive model of the visual future (Zhu et al., 2025; Zhen et al., 2026). This joint objective can expose the policy to interaction dynamics that are absent from action regression alone. Yet the predicted world is usually RGB. Robotic systems also need geometry, semantics, and surface orientation, and recent methods add structured supervision through auxiliary branches or experts (Ma et al., 2026; Yuan et al., 2026a; Zhao et al., 2026). Such designs are effective, but their output vocabulary is fixed by the architecture.

A complementary line of work casts visual understanding as image or video generation. Painter uses images to represent both task demonstrations and outputs (Wang et al., 2023); Vision Banana and GenCeption instruction-tune large generators to emit depth, segmentation, normals, and other visual measurements (Gabeur et al., 2026; Wang et al., 2026). SUV applies a shared video expert to future appearance, semantics, depth, and dynamics in driving, while retaining a separate action expert (Yuan et al., 2026b). These results suggest that the output representation, rather than a task-specific decoder, can identify a visual task.

Action Images (Zhen et al., 2026) makes this idea directly applicable to manipulation: a continuous end-effector trajectory is projected into each camera and rendered as a three-channel Gaussian video. Actions and observations can consequently be denoised in a single video latent sequence. We extend that interface rather than adding prediction heads. In

CoGen, a sampled template specifies which modalities occupy the sequence; atomic tags textualize that template; and a conditioning mask specifies which latent positions are clean. All target positions share the same VAE, denoiser, final projection, scheduler, and loss.

The distinction between template and conditioning is important. For example, video+depth under full RGB conditioning is depth estimation, whereas the same template with only initial frames conditioned is a joint RGB–depth world model. Conversely, changing a tag without changing the segment layout or its target anchor is outside the training support. We therefore use prompt-routed to mean that a prompt-visible task template selects a trained route, not that text has been causally isolated as the only routing signal.

Our contributions are:

- a shared output formulation for RGB, depth, scene-role segmentation, normals, and actions, requiring no learned modality-specific head or modality-specific loss;
- a factorization of each example into a prompt-serialized modality template and an orthogonal conditioning plan, covering joint generation, cross-view generation, inverse prediction, single-frame policy learning, and video-only learning;
- an auditable arm6 training distribution that preserves prompt–layout–target agreement and exact view/time correspondence across all streams; and
- a reproducible, deliberately preliminary validation of the arm6 checkpoint, together with explicit boundaries around the experiments that have not yet been run.

2 Related Work

World-action generation. Unified World Models couple video and action diffusion while retaining stream-specific noise levels (Zhu et al., 2025). Action Images instead renders actions into multiview images and uses one video-generative process (Zhen et al., 2026); this is our starting checkpoint and representation. Camera conditioning follows the same Plücker-ray construction, related to camera-controlled video generation (Sitzmann et al., 2021; Bai et al., 2025).

Structured future supervision. GeoSem-WAM adds geometry and semantic prediction branches to an RGB future objective (Ma et al., 2026). DreamWAM jointly denoises RGB and motion and uses gated residual branches for geometry and semantics (Yuan et al., 2026a). RynnWorld-4D co-generates RGB, depth, and flow with a tri-branch model and obtains actions from an inverse-dynamics head (Zhao et al., 2026). SUV is closer to our visual design: a shared video expert emits several future scene streams without stream-specific visual heads, although action remains a separate expert (Yuan et al., 2026b). CoGendiffers in placing action and every visual modality behind the same generative projection.

Understanding through generation. Generalist vision models unify heterogeneous targets by representing them as images (Wang et al., 2023; Zhuang et al., 2025). Recent image and video generators scale this approach to broad task mixtures selected by instructions (Gabeur et al., 2026; Wang et al., 2026). Our focus is narrower and embodied: synchronized future RGB, geometric, semantic, and action streams are packed into the latent sequence of a WAM, and the conditioning pattern is part of the task definition.

3 Method

3.1 Overview: one ambient space and one head

Let

$$\mathcal{M} = \{\text{video, depth, seg, normal, action}\}. \quad (1)$$

For each modality m , a deterministic codec c_m produces a three-channel clip in $[-1, 1]$. RGB uses the identity codec. Depth, segmentation, and normals use the codecs in section 3.6;

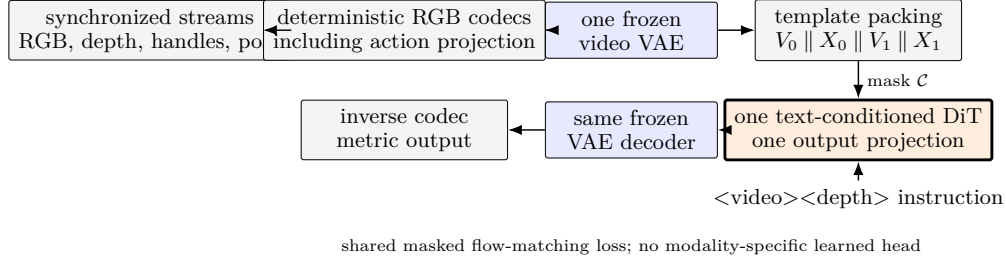


Figure 1: CoGen converts all outputs to an RGB-like video interface. A task template fixes the packed layout and is written into the prompt. The only learned predictor is the shared DiT; the text encoder and video VAE are frozen. X is action, depth, scene-role segmentation, or normal.

Table 1: The arm6 task menu. V, D, S, N, A denote RGB, depth, scene-role segmentation, normal, and action. Template choice, prompt prefix, and packed target are constructed together.

Template	Probability	Prompt prefix	Two-view layout
video+action	0.40	<video><action>	$V_0 \parallel A_0 \parallel V_1 \parallel A_1$
video+depth	0.20	<video><depth>	$V_0 \parallel D_0 \parallel V_1 \parallel D_1$
video+segmentation	0.20	<video><scene-seg>	$V_0 \parallel S_0 \parallel V_1 \parallel S_1$
video+normal	0.20	<video><normal>	$V_0 \parallel N_0 \parallel V_1 \parallel N_1$

actions are projected and rendered using camera calibration. The same frozen causal video VAE E maps every clip to

$$\mathbf{z}_{v,m} = E(c_m(\mathbf{x}_{v,m})) \in \mathbb{R}^{C \times T_z \times H_z \times W_z}, \quad (2)$$

where v indexes a camera view. Thus modality identity is never represented by choosing a decoder head. It is represented by the sampled sequence template, its prompt tags, and the target pixels. The frozen VAE decoder and the inverse codec c_m^{-1} recover the requested measurement at inference.

3.2 Prompt-routed task templates

A template $\Pi \subseteq \mathcal{M}$ specifies the modalities present in an example. Modalities are sorted in the fixed order

$$\text{video} \prec \text{depth} \prec \text{seg} \prec \text{normal} \prec \text{action}, \quad (3)$$

and segments are packed view-major and modality-minor:

$$\mathbf{z}^\Pi = \left\| \big\|_{v=0}^{V-1} \big\|_{m \in \text{ord}_\prec(\Pi)} \mathbf{z}_{v,m}, \quad (4)$$

where $\text{ord}_\prec$ sorts by equation 3 and $\parallel$ denotes concatenation along latent time. With two views, video+depth becomes $V_0 \parallel D_0 \parallel V_1 \parallel D_1$. The corresponding prompt is the bare instruction prefixed by atomic tags, e.g., <video><depth> slide the bottom drawer open. Scene-role segmentation uses the single atomic tag <scene-seg>; its palette is fixed globally, so no episode-specific color map is needed. During training, punctuation is scrubbed by the inherited text path and text is dropped with probability 0.05 for classifier-free guidance.

Arm6 uses only the four two-modality templates in Table 1. These templates all contain four full-length segments and therefore have the same latent-sequence length. Action dropout additionally produces a two-segment video-only example. The generic assembler can express depth+action and video+depth+action, but neither template occurs in arm6: depth and action are trained in separate examples, never jointly in the same packed sequence.

Table 2: Conditioning plans. I, F, and 1 denote initial-latent conditioning, full-segment conditioning, and a clean one-latent segment, respectively. The single-frame policy plan is action-specific.

Name	Segment states	Interpretation
IIII	I-I-I-I	Jointly generate both streams and both views from their initial latent frames.
FIII	F-I-I-I	Condition on the complete first visual view; generate the remaining segments.
FIFI	F-I-F-I	Condition on both complete primary-visual streams; infer the paired stream.
policy	1-I-1-I	Keep one clean frame per RGB view and generate full action segments.
video-only	I-I	After action dropout, generate the two RGB views from initial frames.

3.3 Conditioning plans

The template specifies what is present; a conditioning plan specifies what is given. For every segment, F marks the full segment as clean, while I marks only its first latent frame as clean. Clean positions form a Boolean set $\mathcal{C}$ and are excluded from the loss. A 1 segment is truncated to one latent frame, which is itself clean. For a two-view template $V_0 \parallel X_0 \parallel V_1 \parallel X_1$, the implemented plans are:

For video+action, FIFI is inverse action prediction from observed video. For video+depth, it is video-to-depth estimation. Under IIII, both are generative world models. The assembler tracks cumulative offsets, so full and one-frame segments can coexist, and asserts $\mathcal{C}^c \neq \emptyset$; a fully conditioned example would otherwise consume a training step with zero supervised positions.

3.4 Shared masked flow matching

Let $\epsilon \sim \mathcal{N}(0, I)$ and let σ_t be the scheduler noise level. The full packed sequence is noised as

$$\mathbf{z}_t = (1 - \sigma_t)\mathbf{z}^\Pi + \sigma_t\epsilon, \quad \mathbf{u} = \epsilon - \mathbf{z}^\Pi. \quad (5)$$

After noising, clean positions are restored: $\tilde{\mathbf{z}}_{t,i} = \mathbf{z}_i^\Pi$ for $i \in \mathcal{C}$ and $\tilde{\mathbf{z}}_{t,i} = \mathbf{z}_{t,i}$ otherwise. The text-conditioned DiT f_θ receives the packed latents, the timestep, and VAE-encoded Plücker camera rays. It is optimized by

$$\mathcal{L}(\theta) = \mathbb{E}_{\Pi, t, \epsilon, \mathcal{C}} \left[w_t \frac{\sum_i \mathbf{1}[i \notin \mathcal{C}] \|f_\theta(\tilde{\mathbf{z}}_t, t, \mathbf{c}_{\text{cam}}, \mathbf{c}_{\text{txt}})_i - \mathbf{u}_i\|_2^2}{\sum_i \mathbf{1}[i \notin \mathcal{C}]} \right]. \quad (6)$$

There is no modality-dependent term or weight in equation 6. Camera latents for a view are repeated for every modality segment from that view and injected by the inherited per-block camera encoders. The text encoder and VAE remain frozen; the denoiser, including its one final output projection, is fine-tuned.

3.5 Exact arm6 mixture

The nominal template distribution is given in Table 1. A video+action draw first drops action with probability 0.10. Conditional on retaining action, the inherited sampling logic selects policy with probability 0.10; otherwise it selects IIII/FIII/FIFI with probabilities 0.90/0.05/0.05. Perception templates use preset M2, directly selecting IIII/FIII/FIFI with 0.90/0.05/0.05. The resulting global distribution is

$$p(\text{IIII}, \text{FIII}, \text{FIFI}, \text{policy}, \text{video-only}) = (0.8316, 0.0462, 0.0462, 0.0360, 0.0400). \quad (7)$$

The derivation is in Appendix B. Sampling the template and action dropout in the dataset ensures that the tags, packed layout, and available targets agree before the forward pass.

3.6 Modality codecs

Action images. An action is $\mathbf{a}_t = (\mathbf{p}_t, R_t, g_t)$: 3D position, orientation, and gripper openness. For view v with projection π_v , the renderer projects the gripper center and two tool-frame axis tips,

$$\mathbf{q}_r = \pi_v(\mathbf{p}_t), \quad \mathbf{q}_g = \pi_v(\mathbf{p}_t + \ell R_t \mathbf{e}_x), \quad \mathbf{q}_b = \pi_v(\mathbf{p}_t - \ell R_t \mathbf{e}_z), \quad \ell = 0.1 \text{ m}, \quad (8)$$

and draws one Gaussian per channel,

$$A_{t,c}^v(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x} - \mathbf{q}_c\|_2^2}{2s^2}\right), \quad s = 0.05 \min(H, W). \quad (9)$$

When the gripper is open, weak pixels in the blue channel are floored at 0.25. At decoding, channel peaks from two views define rays whose multiview agreement recovers the 3D pose and orientation (Zhen et al., 2026). Rendering occurs inside the forward pass, so the same 3D trajectory is reprojected through the exact cameras selected for the RGB sample.

Metric depth. Depth d is compressed with

$$y = 1 - \left(1 - \frac{d}{\lambda c}\right)^{\lambda+1}, \quad \lambda = -3, \quad c = 10/3, \quad (10)$$

then linearly embedded along the seven-edge RGB-cube path black–red–yellow–green–cyan–blue–magenta–white (Gabeur et al., 2026). Decoding projects RGB onto the nearest path segment and inverts equation 10. Values outside the strict 0.05–10 m engineering range use an off-path sentinel and decode as invalid rather than as plausible depth.

Surface normals. Normals are derived from synchronized metric depth and camera intrinsics, without additional annotation. For pixel coordinates (u, v) ,

$$\mathbf{n} \propto \left(-\frac{f_x}{z} \frac{\partial z}{\partial u}, -\frac{f_y}{z} \frac{\partial z}{\partial v}, 1\right)^\top, \quad c_{\text{normal}}(\mathbf{n}) = \frac{\mathbf{n} + 1}{2}. \quad (11)$$

Focal lengths are scaled when depth is resized. The decoder maps RGB back to $[-1, 1]^3$ and renormalizes each vector.

Dense scene roles. Raw CoppeliaSim handle maps are converted by a per-episode handle-to-role lookup and a fixed palette. The semantic roles are background, target, goal, robot arm, gripper, distractor, tool, and fixture; unknown is a reserved debugging label that is required to be absent from training targets. Instruction resolution promotes only manipulable distractors to target, while preserving goal and tool identities. Compared with instruction-referring targets, the scene-role maps expose substantially denser supervision: across the 16 tasks, the cross-task median target coverage is 0.97% for referring masks and 12.19% for non-background scene roles, a $12.6\times$ increase. The tag is the atomic `<scene-seg>` because the role palette is global.

4 Experimental Protocol and Current Evidence

4.1 Data and training configuration

We use a self-generated RL Bench dataset (James et al., 2020) with Colosseum-style visual and camera randomization (Pumacay et al., 2024). It contains 1,028 episodes from 16 tasks, four calibrated views, and synchronized RGB, metric depth, object-handle masks, camera parameters, and 7-DoF end-effector states. Training uses the 788 variation-0 episodes. Variations 1 and 2 contain 130 and 110 episodes, respectively, and form a 240-episode held-out set that has not yet been evaluated. Each example samples two of the four views and 41 frames at native stride 3, spanning 121 simulator steps (6.0 s at approximately 6.7 Hz).

We warm-start from the released Action Images checkpoint and fine-tune Wan2.2-TI2V-5B (Wan Team, 2025). The actual arm6 run uses 512×512 inputs, two GPUs, per-device

Table 3: Preliminary arm6 diagnostic from one held-in episode at step 4,500. The same checkpoint is invoked with a different template prefix for each row. Best values across conditioning modes are not emphasized because the modes expose different amounts of information.

Prompt route	Metric	III	FIII	FIFI
<video><action>	position med. (cm) ↓	5.40	4.50	1.85
	rotation med. (°) ↓	20.89	19.81	18.73
<video><depth>	AbsRel ↓	0.160	0.069	1.335
<video><scene-seg>	macro-mIoU ↑	0.373	0.355	0.494
<video><normal>	mean cosine ↑	0.904	0.937	0.952

batch size 1, gradient accumulation 2 (effective global batch 4), bf16, full-denoiser fine-tuning, gradient checkpointing, and ZeRO-2 with CPU-offloaded optimizer state. Adam uses learning rate 5×10^{-7} , 1,000 warm-up steps, then a constant schedule. The seed is 42. The text encoder and VAE are frozen. The current checkpoint is step 4,500 of a planned 10,000; the run is therefore not treated as converged.

4.2 Metrics and evaluation scope

Depth is decoded to meters and scored by AbsRel. Scene-role segmentation uses macro-IoU over roles present in the ground truth. Normals use mean cosine similarity. Actions are decoded from the two-view heatmaps and scored by median 3D position error and geodesic rotation error. III is the principal generative regime; FIII, FIFI, and policy diagnose other conditioning plans that occur during training.

All results in Table 3 use the same arm6 step-4,500 weights, the same held-in episode (open_drawer/variation0/episode0), resolution, temporal stride, and random seed. They validate routing, decoding, and metric plumbing; $n = 1$ does not support a generalization claim or a comparison to prior methods.

The diagnostic establishes a limited but necessary fact: all four routes produce outputs that survive their inverse codecs and yield non-degenerate task metrics from one shared checkpoint. The information ordering is reflected in the action row: observing two full RGB clips (FIFI) is easier than jointly imagining RGB and action (III). The depth FIFI result moves in the opposite direction and is unstable on this episode. This mode receives only 4.62% of all arm6 samples, shared across perception and action templates; the single example cannot distinguish under-training from episode-specific failure. Under policy, which receives only single RGB frames, action position and rotation errors are 14.79 cm and 26.93° on the same episode.

As a small cross-episode check, III depth AbsRel at step 4,500 is 0.160, 0.065, 0.116, and 0.059 on one held-in episode each from open_drawer, push_buttons, stack_wine, and turn_tap (mean 0.100). We keep this observation in prose because four training-variation episodes are still a pipeline check, not the planned held-out sweep.

5 Discussion and Limitations

The routing signal is not isolated. Every predicted segment receives a clean first latent frame. For a depth target, for example, that anchor is already a depth-encoded image. The modality tag, sequence position, and target anchor therefore agree throughout training. The present results establish switching through the complete template interface, not switching caused by text alone. We deliberately do not report f0f0: it removes the target anchors, is absent from training, and would be an out-of-distribution test rather than a clean prompt ablation. A valid causal test requires adding tag-only examples to the training distribution.

No joint structured-action target. Arm6 trains video+action and each video+structured-modality template separately. It has not trained depth+action or video+depth+action.

Consequently, the current checkpoint does not establish that depth and action can be generated jointly or that separately sampled routes are mutually consistent. A three-modality template is supported by the assembler but increases sequence length from four to six segments.

Incomplete evaluation. The principal run has reached 4,500 of 10,000 planned steps and uses one seed. The 240 held-out variation episodes, closed-loop rollouts, matched action-only and per-modality controls, and checkpoint selection study have not been completed. We therefore avoid claims about task generalization, policy success, interference, or superiority to specialized heads. Results from other exploratory arms are not used as evidence for arm6.

Simulation supervision and consistency. Depth, handles, and calibration are simulator ground truth. Real-data extension will require sensor depth or pseudo-labels whose metric scale and temporal consistency are verified. Moreover, a shared head and shared latent space do not guarantee physical agreement between outputs from separate stochastic generations; explicit joint templates or cross-modal consistency objectives would be needed to test that property.

6 Conclusion

CoGen treats modality as part of a WAM’s generative interface rather than as an architectural branch. Deterministic RGB-like codecs place video, actions, depth, scene roles, and normals in the same frozen-VAE space; a prompt-visible task template specifies their packed layout; a separate mask specifies what is observed; and one DiT projection is trained with one masked flow-matching objective. The arm6 diagnostic confirms that this pipeline can route four auxiliary outputs through one checkpoint. The next empirical requirement is clear: finish arm6, evaluate the held-out variations and closed loop, and compare against matched controls without introducing untrained prompt-only or depth-action conditions.

Reproducibility Statement

The launcher records the template mix, conditioning preset, segmentation protocol, dataset, resolution, temporal stride, seed, and initialization in the output path or run configuration. The dataset records the sampled frame indices, view indices, and view directories and uses that provenance for every modality. Startup checks validate requested modality availability, segmentation coverage, absence of the reserved unknown role, and non-empty prediction masks. Exact commands and artifact paths for the reported diagnostics are listed in paper/README.md.

References

- Jianhong Bai, Menghan Fan, Xiaoyu Xia, Xiao Bai, Ziyang Lu, Xintao Wang, Yujiu Zhang, Hongtao Yang, Pengfei Wan, and Di Zhang. ReCamMaster: Camera-controlled generative rendering from a single video. arXiv preprint arXiv:2503.11647, 2025.
- Valentin Gabeur, Shangbang Long, Songyou Peng, Paul Voigtlaender, Shuyang Sun, Yanan Bao, Karen Truong, Zhicheng Wang, Wenlei Zhou, Jonathan T. Barron, Kyle Genova, Nithish Kannan, Sherry Ben, Yandong Li, Mandy Guo, Suhas Yogin, Yiming Gu, Huizhong Chen, Oliver Wang, Saining Xie, Howard Zhou, Kaiming He, Thomas Funkhouser, Jean-Baptiste Alayrac, and Radu Soricut. Image generators are generalist vision learners. arXiv preprint arXiv:2604.20329, 2026.
- Stephen James, Zicong Ma, David Rovick Arrojo, and Andrew J. Davison. RL Bench: The robot learning benchmark & learning environment. IEEE Robotics and Automation Letters, 5(2):3019–3026, 2020.
- Fulong Ma, Daojie Peng, Wenjun Yue, Jiahang Cao, Bintao Wang, Qiang Zhang, and Jun Ma. GeoSem-WAM: Geometry- and semantic-aware world action models. arXiv preprint arXiv:2606.03188, 2026.

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- Wilbert Pumacay, Ishika Singh, Jiafei Duan, Ranjay Krishna, Jesse Thomason, and Dieter Fox. The colosseum: A benchmark for evaluating generalization for robotic manipulation. In *Robotics: Science and Systems*, 2024.
- Vincent Sitzmann, Semon Rezhikov, William T. Freeman, Joshua B. Tenenbaum, and Fredo Durand. Light field networks: Neural scene representations with single-evaluation rendering. In *Advances in Neural Information Processing Systems*, 2021.
- Wan Team. Wan: Open and advanced large-scale video generative models. arXiv preprint arXiv:2503.20314, 2025.
- Letian Wang, Chuhan Zhang, Rishabh Kabra, Jasper Uijlings, Steven Waslander, Andrew Zisserman, Joao Carreira, Kaiming He, Misha Andriluka, Eduard Gabriel Bazavan, Andrei Zanfir, and Cristian Sminchisescu. Video generation models are general-purpose vision learners. In *European Conference on Computer Vision*, 2026.
- Xinlong Wang, Wen Wang, Yue Cao, Chunhua Shen, and Tiejun Huang. Images speak in images: A generalist painter for in-context visual learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Shanglin Yuan, Weiheng Zhao, Xin Shi, Haoyi Jiang, Xianda Guo, Liu Liu, Wenyu Liu, Wei Sui, and Xinggang Wang. DreamWAM: Beyond RGB future prediction for world action models. arXiv preprint arXiv:2608.04996, 2026a.
- Yibo Yuan, Jiacheng Fu, Jiangtong Zhu, Yi Li, Jianhua Han, Meng Tian, Zhuohan Liu, Zhiwei Xiong, Hang Xu, Jianwu Fang, and Jianru Xue. SUV: Future scene understanding as video generation for end-to-end driving. arXiv preprint arXiv:2608.03084, 2026b.
- Haoyu Zhao, Xingyue Zhao, Siteng Huang, Xin Li, Deli Zhao, and Zhongyu Li. RynnWorld-4D: 4d embodied world models for robotic manipulation. arXiv preprint arXiv:2607.06559, 2026.
- Haoyu Zhen, Zixian Gao, Qiao Sun, Yilin Zhao, Yuncong Yang, Yilun Du, Pengsheng Guo, Tsun-Hsuan Wang, Yi-Ling Qiao, and Chuang Gan. Action images: End-to-end policy learning via multiview video generation. arXiv preprint arXiv:2604.06168, 2026.
- Chuning Zhu, Raymond Yu, Siyuan Feng, Benjamin Burchfiel, Paarth Shah, and Abhishek Gupta. Unified world models: Coupling video and action diffusion for pretraining on large robotic datasets. arXiv preprint arXiv:2504.02792, 2025.
- Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, Jingtao Li, Jiabo Huang, Vikash Sehwal, Vivek Sharma, Hirotaka Shinozaki, Felan Carlo Garcia, Yihao Zhan, Naohiro Adachi, Ryoji Eki, Michael Spranger, Peter Stone, and Lingjuan Lyu. Argus: A compact and versatile foundation model for vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

A Codec and data-integrity details

Depth inversion. For an RGB value γ , decoding searches the seven RGB-cube edges with endpoints $(\mathbf{v}_k, \mathbf{v}_{k+1})$:

$$(k^*, \alpha^*) = \arg \min_{\substack{k \in \{0, \dots, 6\} \\ \alpha \in [0, 1]}} \|\gamma - [(1 - \alpha)\mathbf{v}_k + \alpha\mathbf{v}_{k+1}]\|_2^2, \quad \hat{y} = \frac{k^* + \alpha^*}{7}. \quad (12)$$

The inverse is $\hat{d} = \lambda c [1 - (1 - \hat{y})^{1/(\lambda+1)}]$. Pixels whose squared distance to the path exceeds 0.15, or whose recovered depth is outside the valid interval, decode to NaN. Depth is resized with nearest-neighbor interpolation before encoding so that no metric values are synthesized by the resize operation.

Scene-role palette. The fixed sRGB colors are background (0,0,0), target (230,25,75), goal (60,180,75), robot arm (0,130,200), gripper (70,240,240), distractor (255,225,25), tool (240,50,230), fixture (145,30,180), and reserved unknown (245,130,48). Decoding is nearest-palette classification restricted to roles that can legally appear in the episode. It does not prune small connected components because several valid target objects occupy only a few pixels. The CoppeliaSim no-object value, truncated to 0xFFFF in the stored mask, maps to background rather than unknown.

Frozen-VAE checks. Stored admission checks provide two useful upper-bound measurements. On 12 episodes, normal clips round-tripped through the frozen VAE at mean cosine 0.9906 (mean angular error 7.88°). On 48 episodes evaluated at 256^2 , scene-role targets had median IoU 0.965 after the VAE round-trip; the RGB control PSNR was 29.59 dB. These checks validate representational survival, not learned prediction quality. We do not report an unstored numeric VAE gate for depth or action.

Alignment invariants. Every perception stream is loaded with the exact frame indices and view directories chosen by the RGB loader. Each camera’s Plücker embedding is then duplicated for all modality segments from that camera. Requested templates unsupported by a dataset fail at startup rather than being silently shortened. For scene roles, all 788 training episodes have target metadata; startup probes find no reserved-unknown ground-truth pixels.

B Realized arm6 probabilities

Table 4: Probability decomposition for a nominal video+action draw. These probabilities sum to one within that template before multiplying by its arm6 weight of 0.4.

Outcome	Composition	Within-template probability
video-only	0.10	0.1000
policy	0.90×0.10	0.0900
III	$0.90 \times 0.90 \times 0.90$	0.7290
FII	$0.90 \times 0.90 \times 0.05$	0.0405
FIFI	$0.90 \times 0.90 \times 0.05$	0.0405

Perception templates jointly carry probability 0.6 and use M2, contributing 0.54 to III and 0.03 each to FII and FIFI. The action template contributes 0.2916, 0.0162, 0.0162, 0.036, and 0.04 to III, FII, FIFI, policy, and video-only, respectively. Summing the two sources gives equation 7.

C Condition-mask construction

For a full-length latent segment with temporal length T_z , an I state marks the first latent slice as clean and leaves $T_z - 1$ slices as prediction targets. An F state marks all T_z slices clean. A 1 state first truncates the segment to one slice, then marks that slice clean. Offsets are accumulated from the realized segment lengths rather than computed as multiples of T_z ; this is required for the policy layout $[V_0^1 \| A_0 \| V_1^1 \| A_1]$. The loss mask uses True=clean/given; equation 6 explicitly complements it.

D Evaluation status and number provenance

The route-grid numbers come from reports/modality_mode_grid/arm6_4500/metrics.json. The additional depth measurements come from the three sibling arm6_4500_ep_* directories. Frozen-VAE measurements come from reports/vae_roundtrip_normal.txt and reports/vae_roundtrip_seg.txt. Dataset counts and segmentation-density statistics are re-computed from the active rlbench_selfgen_512_aug tree and reports/seg_pixel_stats.csv, respectively.

Table 5: Status of the empirical cells discussed in the paper. “Done” here means that a stored artifact exists, not that the scope is sufficient for a final benchmark.

Cell	Status	Scope
arm6 step-4,500 route grid	Done	one held-in episode, all trained routes
arm6 depth spot check	Done	four held-in episodes
normal frozen-VAE round-trip	Done	12 episodes
scene-role frozen-VAE round-trip	Done	48 episodes at 256 ²
arm6 held-out variations 1–2	Not done	planned 240 episodes
arm6 closed-loop policy	Not done	no success-rate claim
arm6 matched controls	Not done	no comparative claim
depth+action / video+depth+action	Not trained	excluded from results
prompt-only target without anchor	Not trained	f0f0 excluded

E Reproduction command

The principal run is launched from the repository root with

GPUS=1,2 bash scripts/train_arm.sh arm6

which resolves to

video+action@0.4,video+depth@0.2,
video+segmentation@0.2,video+normal@0.2

with SEG_MODE=scene_roles and PERCEPTION_MASK_MIX=M2. The current run’s logged configuration records gradient accumulation 2; this runtime value, rather than the launcher’s default value, is the one reported in the paper.